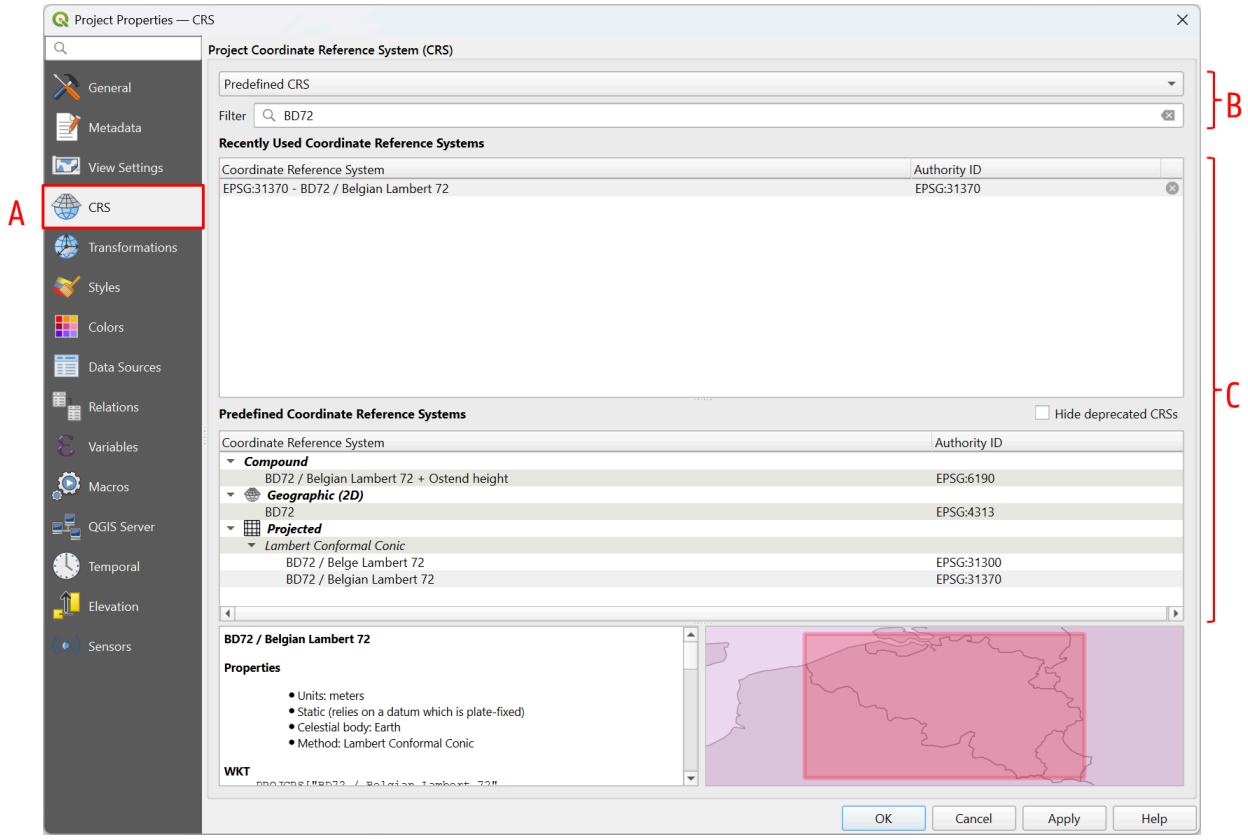
# 5.2.1 Coordinate system of the QGIS project

QGIS offers **thousands of coordinate reference systems** (CRSs) to choose from. As you have learned on the pages on geographic and projected coordinate systems, the choice of CRS matters: different systems represent the Earth's surface in different ways. In QGIS, this is not just a theoretical issue. Your choice affects how your map looks and how spatial information is represented.

## Changing the coordinate system

To change the coordinate system of your project, choose *Project* ▸ *Properties…*. You can also click the CRS code at the right end of the status bar (*EPSG:31370*, for example).

In the *Project Properties* window, open the *CRS* tab (A).
Select the coordinate system you want in the list (C). If it is not visible, type its name or ID in the **Filter** box (B).




> **Tip**
>
> Most coordinate systems have a **unique code**, the EPSG code. Choosing a system by its code is safer than choosing it by its name: some systems have almost the same name but are **not interchangeable**, for example *BD72 / Belgian Lambert 72* (EPSG:31370) and *BD72 / Belge Lambert 72* (EPSG:31300, an older variant with slightly different parameters).

Some widely used **projected systems** include:

- For Belgium: *BD72 / Belgian Lambert 72* (EPSG:31370) and the newer *ETRS89 / Belgian Lambert 2008* (EPSG:3812)
- For the Netherlands: *Amersfoort / RD New* (EPSG:28992)
- For France: *RGF93 v1 / Lambert-93* (EPSG:2154)
- For the United Kingdom: *OSGB36 / British National Grid* (EPSG:27700)
- For Europe: *ETRS89-extended / LAEA Europe* (EPSG:3035), an equal-area system
- For the world: *WGS 84 / Pseudo-Mercator* (EPSG:3857), as in Google Maps and OpenStreetMap. It is almost conformal and strongly enlarges areas towards the poles, so **do not use it to compare or measure areas**.
- For a region anywhere in the world: a *UTM zone*. In the UTM (Universal Transverse Mercator) system, the world is divided into 60 zones of 6° of longitude, each with its own projection that has little distortion within the zone. Every zone has two coordinate systems: one for the northern hemisphere (N) and one for the southern hemisphere (S). For Belgium, the most suitable is zone 31N (in QGIS: *WGS 84 / UTM zone 31N*, EPSG:32631).

The most widely used **geographic systems** are *WGS 84* (EPSG:4326) for the world, as used by GPS, and *ETRS89* (EPSG:4258) for Europe.

On epsg.io you can look up a coordinate system by its code, its name or a country. For each system, the website shows the area it is meant for, its unit and its parameters.

# Projected versus geographic CRS in QGIS

## Projected

The projected coordinate system you choose largely determines what your map looks like, as explained on the page on projections.

- Select a coordinate system that is **suitable for the region** you want to map. Otherwise, the distortions on your map can be large.
- Select a coordinate system whose **distortion properties suit the purpose of the map**. If area comparisons are important, an equal-area system is a good choice. For a general-purpose map, a conformal or compromise projection may be more suitable.

## Geographic

In theory, setting a geographic coordinate system for your QGIS project poses a challenge. Geographic coordinates describe positions on a curved reference ellipsoid, while the QGIS map canvas is a flat, two-dimensional plane.

**That is why QGIS also applies a projection to geographic coordinate systems for display.** By default, this is a *cylindrical equidistant projection* (plate carrée). This affects how the data are displayed in the map canvas, but it does not change the underlying coordinates of your data.

> Tip
>
> In practice, geographic coordinate systems are **usually not well suited for displaying** geodata in QGIS or on a map, but they are **well suited for storing** spatial information. For map display, choose a projected coordinate system that suits your region and the purpose of your map.